

Year 7 • Lesson 10.06 The median and range: answers (teacher copy)

10.06 The median and range (booklet p. 3)

Warm-up

1. 2, 4, 6, 7, 9
2. 8
3. 15
4. 11

WE DO examples

1. Ex 1: 3, 4, 4, 5, 7, 8, 9 → 5
2. Ex 2: $20 - 5 = 15$
3. Ex 3: median 7.5, range 9
4. Ex 4: median 35, range 18
5. Ex 5: median 12, range 33, mean 17.33; the median, because 44 is an outlier
6. Ex 6: e.g. 3, 5, 8, 9, 13

Set A

1. 3
2. 4
3. 5
4. 6
5. 11
6. 7
7. 14
8. 3
9. 22
10. 7

Set B

1. 7
2. 17
3. 5
4. 12
5. 15
6. 0
7. 38
8. 4.3
9. 15
10. 7

Set C

1. median 5.5, range 7
2. median 14.5, range 9
3. median 3.1, range 3.6
4. median 7, range 9
5. median 1, range 9
6. median 22, range 5
7. median 2, range 3
8. median 151, range 15
9. median 175, range 200
10. median 1, range 3

Set D

1. e.g. 2, 4, 6, 8, 10
2. $7 + 15 = 22$
3. Median $7 \rightarrow 7.5$; range $6 \rightarrow 16$
4. Class B is more consistent (smaller range); Class A did better overall (higher median)
5. The mean uses every value's size, but the median only depends on the middle position
6. 10 and 10
7. e.g. 3, 6, 8, 10, 13
8. 195 cm
9. Yes, when every value is the same
10. median 12.5, range 41. Not really: no one is aged about 12.5

Extension

1. The 20th and 21st
2. 20 values: 10th and 11th are both 3, so the median is 3
3. 50.5

Exit ticket

1. Level 1: 7
2. Level 2: median 6.5, range 8
3. Level 3: e.g. 2, 4, 6, 9, 12